

## 1. Preprocessing & Windowing

DASIG: 60 Subjects  $\times$  3 Trials  
5 Wireless MIMUs (65 Channels) @ 200 Hz



Sliding Window: 0.5 s (100 steps, 50% overlap)  
Per-Trial Z-Score Standardization



Window-Level Labeling:  $y = \max(t\_labels)$   
69,094 Windows (87.6% SAFE / 12.4% DANGER)



## 2. Dynamic Kinematic Feature Expansion

**Kinematic Channels ( $3 \times 65 = 195$ )**

Position (Raw):  $X \in \mathbb{R}^{65 \times 100}$



Velocity (1st Deriv):  $V = dX/dt \in \mathbb{R}^{65 \times 100}$



Acceleration (2nd Deriv):  $A = d^2X/dt^2 \in \mathbb{R}^{65 \times 100}$



**Expanded Kinematic Tensor: [X; V; A]**  
**Dimensions: (B, 195, 100)**